

Módulo 1 — Práctica guiada: despliegue de un servidor nginx en

Terminal

```
Windows PowerShell
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PS C:\Users\maria> cd ..
PS C:\Users> cd ..
PS C:\> cd Sena\Materias5Trimestre\Docke
PS C:\Sena\Materias5Trimestre\Docke> docker version --
Client:
Version:           29.6.2
API version:       1.55
Go version:        go1.26.5
Git commit:        dfc4efb
Built:             Thu Jul 16 16:14:59 2026
OS/Arch:           windows/amd64
Context:           desktop-linux

Server: Docker Desktop 4.83.0 (234302)
Engine:
Version:           29.6.2
API version:       1.55 (minimum version 1.40)
Go version:        go1.26.5
Git commit:        3d80467
Built:             Thu Jul 16 16:12:20 2026
OS/Arch:           linux/amd64
Experimental:      false
containerd:
Version:           v2.2.5
```

contenedor

docker run -d --name practica1 -p 8081:80 nginx

```
PS C:\Sena\Materias5Trimestre\Docke> docker run -d --name practica1 -p 8081:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
f340c1b7c1d6: Pull complete
956faab5efb3: Pull complete
c12f394dea35: Pull complete
02fc02c4ab8d: Pull complete
07db7bf2649b: Pull complete
6310eb16bf42: Pull complete
a44b5c8be616: Pull complete
76f27c02d218: Download complete
25202a7045eb: Download complete
Digest: sha256:05b8cb60c354a44ab824ea6e7dc69b46d50762cdbe728a347a5b656e6fb3d7c4
Status: Downloaded newer image for nginx:latest
16fa1488d42ab4b66482f521b0afd42e8f183b7454c20d20fcf370c11e44c690
PS C:\Sena\Materias5Trimestre\Docke> docker run -it --name consola ubuntu:24.04 bash
Unable to find image 'ubuntu:24.04' locally
24.04: Pulling from library/ubuntu
76f27c02d218: Download complete
76f27c02d218: Download complete
25202a7045eb: Download complete
Digest: sha256:05b8cb60c354a44ab824ea6e7dc69b46d50762cdbe728a347a5b656e6fb3d7c4
Status: Downloaded newer image for nginx:latest
```

Módulo 2 — Práctica guiada: modos de ejecución de contenedores (interactivo vs. detached)

2.1 Modo interactivo (-it): contenedor consola con Ubuntu 24.04

```
PS C:\Sena\Materias5Trimestre\Docker> docker run -it --name consola ubuntu:24.04 bash
Unable to find image 'ubuntu:24.04' locally
24.04: Pulling from library/ubuntu
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
root@a5a2bbf7940b:/# exit
exit
PS C:\Sena\Materias5Trimestre\Docker>
```

2.2 Modo detached (-d): contenedor web con nginx y revisión de logs

```
PS C:\Sena\Materias5Trimestre\Docker> docker run -d --name web -p 8080:80 nginx
1a72e5f0e3fb25a1e2b2cb14dc40a46e995a5f7c2f027574fb7a5c837fd1219
PS C:\Sena\Materias5Trimestre\Docker> docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
1a72e5f0e3fb	nginx	"/docker-entrypoint..."	18 seconds ago	Up 17 seconds	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	web
16fa1488d42a	nginx	"/docker-entrypoint..."	9 minutes ago	Up 9 minutes	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp	practica1

```
PS C:\Sena\Materias5Trimestre\Docker>
```

```
> ^C
PS C:\Sena\Materias5Trimestre\Docker> docker logs -f web
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
2026/09/03 18:00:18 [notice] 1#1: start worker process 34
2026/09/03 18:00:18 [notice] 1#1: start worker process 35
2026/09/03 18:00:18 [notice] 1#1: start worker process 36
2026/09/03 18:00:18 [notice] 1#1: start worker process 37
2026/09/03 18:00:18 [notice] 1#1: start worker process 38
2026/09/03 18:00:18 [notice] 1#1: start worker process 39
```

 Herramienta

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PS C:\Sena\Materias5Trimestre\Docke> docker logs -f web
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
2026/09/03 18:00:18 [notice] 1#1: start worker process 34
2026/09/03 18:00:18 [notice] 1#1: start worker process 35
2026/09/03 18:00:18 [notice] 1#1: start worker process 36
2026/09/03 18:00:18 [notice] 1#1: start worker process 37
2026/09/03 18:00:18 [notice] 1#1: start worker process 38
2026/09/03 18:00:18 [notice] 1#1: start worker process 39
2026/09/03 18:00:18 [notice] 1#1: start worker process 40
172.17.0.1 - - [03/Sep/2026:18:01:44 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,
like Gecko) Chrome/152.0.0.0 Safari/537.36" "-"
2026/09/03 18:01:45 [error] 30#30: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, se
rver: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://localhost:8080/"
172.17.0.1 - - [03/Sep/2026:18:01:45 +0000] "GET /favicon.ico HTTP/1.1" 404 555 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64;
x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.0.0 Safari/537.36" "-"

```

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.

To engage with the community please visit community.nginx.org.

For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

2.4 Acceso a contenedores en marcha con docker exec

Ejemplo 1 — abrir una terminal dentro del nginx en marcha

```

PS C:\Sena\Materias5Trimestre\Docke> docker exec -it web bash
root@1a72e5f0e3fb:/# ls /usr/share/nginx/html
50x.html  index.html
root@1a72e5f0e3fb:/# exit
exit

```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → [docker debug web](#)

Learn more at <https://docs.docker.com/go/debug-cli/>

```
PS C:\Sena\Materias5Trimestre\Docke>
```

Ejemplo 2 — una sola orden, sin quedarte dentro

```
PS C:\Sena\Materias5Trimestre\Docker> docker exec web nginx -v
nginx version: nginx/1.31.5
PS C:\Sena\Materias5Trimestre\Docker>
```

2.5 Variables de entorno y límites de recursos

```
PS C:\Sena\Materias5Trimestre\Docker> docker run -d --name db -e POSTGRES_USER=admin
-e POSTGRES_PASSWORD=secreto -e POSTGRES_DB=tienda postgres:16
9268384f44e197998f36e4d0242cf200df92a77e2b611cddccec4d7d3c5f2ba
PS C:\Sena\Materias5Trimestre\Docker> docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
9268384f44e1	postgres:16	"docker-entrypoint.s..."	27 seconds ago	Up 26 seconds	5432/tcp	db
1a72e5f0e3fb	nginx	"/docker-entrypoint..."	12 minutes ago	Up 12 minutes	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp	web
16fa1488d42a	nginx	"/docker-entrypoint..."	21 minutes ago	Up 21 minutes	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp	practical

```
PS C:\Sena\Materias5Trimestre\Docker> docker logs db
The files belonging to this database system will be owned by user "postgres".
This user must also own the server process.

The database cluster will be initialized with locale "en_US.utf8".
The default database encoding has accordingly been set to "UTF8".
The default text search configuration will be set to "english".

Data page checksums are disabled.

fixing permissions on existing directory /var/lib/postgresql/data ... ok
creating subdirectories ... ok
selecting dynamic shared memory implementation ... posix
selecting default max_connections ... 100
selecting default shared_buffers ... 128MB
selecting default time zone ... Etc/UTC
creating configuration files ... ok
running bootstrap script ... ok
performing post-bootstrap initialization ... ok

Success. You can now start the database server using:

    pg_ctl -D /var/lib/postgresql/data -l logfile start

initdb: warning: enabling "trust" authentication for local connections
initdb: hint: You can change this by editing pg_hba.conf or using the option -A, or --auth-local and --auth-host, the next time you run initdb
.
waiting for server to start....2026-09-03 18:12:52.906 UTC [49] LOG:  starting PostgreSQL 16.14 (Debian 16.14-1.pgdg13+1) on x86_64-pc-linux-g
nu, compiled by gcc (Debian 14.2.0-19) 14.2.0, 64-bit
2026-09-03 18:12:52.912 UTC [49] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2026-09-03 18:12:52.928 UTC [52] LOG:  database system was shut down at 2026-09-03 18:12:51 UTC
2026-09-03 18:12:52.941 UTC [49] LOG:  database system is ready to accept connections
done
server started
waiting for server to shut down....2026-09-03 18:12:53.381 UTC [49] LOG:  received fast shutdown request
2026-09-03 18:12:53.412 UTC [49] LOG:  aborting any active transactions
2026-09-03 18:12:53.416 UTC [49] LOG:  background worker "logical replication launcher" (PID 55) exited with exit code 1
2026-09-03 18:12:53.421 UTC [50] LOG:  shutting down
2026-09-03 18:12:53.432 UTC [50] LOG:  checkpoint starting: shutdown immediate
2026-09-03 18:12:53.838 UTC [50] LOG:  checkpoint complete: wrote 926 buffers (5.7%); 0 WAL file(s) added, 0 removed, 0 recycled; write=0.055
s, sync=0.329 s, total=0.417 s; sync files=301, longest=0.005 s, average=0.002 s; distance=4273 kB, estimate=4273 kB; lsn=0/191F0E0, redo lsn=
0/191F0E0
2026-09-03 18:12:53.851 UTC [49] LOG:  database system is shut down
done
server stopped

PostgreSQL init process complete; ready for start up.
```

```

2026-09-03 18:12:53.935 UTC [1] LOG:  starting PostgreSQL 16.14 (Debian 16.14-1.pgdg13+1) on x86_64-pc-linux-gnu, compiled by gcc (Debian 14.2
.0-19) 14.2.0, 64-bit
2026-09-03 18:12:53.937 UTC [1] LOG:  listening on IPv4 address "0.0.0.0", port 5432
2026-09-03 18:12:53.937 UTC [1] LOG:  listening on IPv6 address "::", port 5432
2026-09-03 18:12:53.946 UTC [1] LOG:  listening on Unix socket "/var/run/postgresql/.s.PGSQL.5432"
2026-09-03 18:12:53.957 UTC [65] LOG:  database system was shut down at 2026-09-03 18:12:53 UTC
2026-09-03 18:12:53.969 UTC [1] LOG:  database system is ready to accept connections
PS C:\Sena\Materias5Trimestre\Docker>

```

Ejemplo 2 — límites de memoria y CPU

```

waiting for server to shut down...2026-09-03 18:12:53.381 UTC [49] LOG:  received fast shutdown request
2026-09-03 18:12:53.412 UTC [49] LOG:  aborting any active transactions
CONTAINER ID   NAME      CPU %     MEM USAGE / LIMIT   MEM %     NET I/O       BLOCK I/O     PIDS
408796df023f   api       0.00%     11.98MiB / 512MiB    2.34%     872B / 126B    0B / 8.19kB    13

PS C:\Sena\Materias5Trimestre\Docker>

```

2.6 Mapeo de puertos.

```

PS C:\Sena\Materias5Trimestre\Docker> docker run -d --name db -p 5432:5432 -e POSTGRES_PASSWORD=x postgres:16
6d1d6b8a5f43f4533ff0adac61495b72781caf669c3b8a03d0eb2b2f8848d076
PS C:\Sena\Materias5Trimestre\Docker>
PS C:\Sena\Materias5Trimestre\Docker> docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS
PORTS                NAMES
6d1d6b8a5f43   postgres:16 "docker-entrypoint.s..." 53 seconds ago Up 53 seconds
0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp   db
408796df023f   nginx     "/docker-entrypoint...." 14 minutes ago Up 14 minutes
0.0.0.0:3000->80/tcp, [::]:3000->80/tcp       api
1a72e5f0e3fb   nginx     "/docker-entrypoint...." 29 minutes ago Up 29 minutes
0.0.0.0:8080->80/tcp, [::]:8080->80/tcp       web
16fa1488d42a   nginx     "/docker-entrypoint...." 38 minutes ago Up 38 minutes
0.0.0.0:8081->80/tcp, [::]:8081->80/tcp       practica1
PS C:\Sena\Materias5Trimestre\Docker>

```

```

PS C:\Sena\Materias5Trimestre\Docke> docker volume create datos_db
datos_db
PS C:\Sena\Materias5Trimestre\Docke> docker run -d --name db -v datos_db:/var/lib/p
ostgresql/data -e POSTGRES_PASSWORD=x postgres:16
6a4dddc86dfcdf34faf2db39e5b6117a9115f487e47130ad5fbca6e5351e224
PS C:\Sena\Materias5Trimestre\Docke> docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
6a4dddc86df	postgres:16	"docker-entrypoint.s..."	14 seconds ago	Up 13 seconds
408796df023f	nginx	"/docker-entrypoint...."	22 minutes ago	Up 22 minutes
1a72e5f0e3fb	nginx	"/docker-entrypoint...."	37 minutes ago	Up 37 minutes
16fa1488d42a	nginx	"/docker-entrypoint...."	46 minutes ago	Up 46 minutes

```

PS C:\Sena\Materias5Trimestre\Docke>

```

2.7 Volúmenes

Paso A — crear el volumen y el contenedor

```

PS C:\Sena\Materias5Trimestre\Docke> docker rm -f db
db
PS C:\Sena\Materias5Trimestre\Docke> docker volume create datos_db
datos_db
PS C:\Sena\Materias5Trimestre\Docke> docker run -d --name db -v datos_db:/var/lib/p
ostgresql/data -e POSTGRES_PASSWORD=x postgres:16
6a4dddc86dfcdf34faf2db39e5b6117a9115f487e47130ad5fbca6e5351e224
PS C:\Sena\Materias5Trimestre\Docke> docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS
6a4dddc86df	postgres:16	"docker-entrypoint.s..."	14 seconds ago	Up 13 seconds
408796df023f	nginx	"/docker-entrypoint...."	22 minutes ago	Up 22 minutes
1a72e5f0e3fb	nginx	"/docker-entrypoint...."	37 minutes ago	Up 37 minutes
16fa1488d42a	nginx	"/docker-entrypoint...."	46 minutes ago	Up 46 minutes

```

PS C:\Sena\Materias5Trimestre\Docke>

```

Paso B — crear una tabla dentro de la base

```

PS C:\Sena\Materias5Trimestre\Docker> docker exec -it db psql -U postgres
psql (16.14 (Debian 16.14-1.pgdg13+1))
Type "help" for help.

postgres=# CREATE TABLE prueba (id INT, nombre TEXT);
CREATE TABLE
postgres=# INSERT INTO prueba VALUES (1, 'Mariana');
INSERT 0 1
postgres=# SELECT * FROM prueba;
 id | nombre 
----+-----
  1 | Mariana
(1 row)

postgres=# /q
postgres=#

```

2.8 Redes.

Paso A — crear la red

```

PS C:\Sena\Materias5Trimestre\Docker> docker network create mired
b8418fa3178b056a3df2ae03fd65dde4133bc894d5e884262a32f5d10102ec0d
PS C:\Sena\Materias5Trimestre\Docker> docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
b6bfd7fa7a46	bridge	bridge	local
1579d31521e4	host	host	local
ffe4a26094ad	mi-stack-v2_red_stack	bridge	local
45c0126b70fa	minikube	bridge	local
b8418fa3178b	mired	bridge	local
8b647dc6620b	none	null	local

```

PS C:\Sena\Materias5Trimestre\Docker>

```

Paso B — levantar un contenedor dentro de esa red

```

PS C:\Sena\Materias5Trimestre\Docker> docker run -d --name servidor --network mired
nginx
9b2a80dc053b86bfdecc342b7a94482cfdc45eea4a6608259431b6a405805796
PS C:\Sena\Materias5Trimestre\Docker>

```

Paso C — la prueba: otro contenedor le hace ping por nombre


```
PS C:\Sena\Materias5Trimestre\Docker> docker run --rm -it --network mired busybox ping -c 2 servidor
PING servidor (172.19.0.2): 56 data bytes
64 bytes from 172.19.0.2: seq=0 ttl=64 time=9.330 ms
64 bytes from 172.19.0.2: seq=1 ttl=64 time=0.347 ms
```

```
--- servidor ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.347/4.838/9.330 ms
PS C:\Sena\Materias5Trimestre\Docker>
```

```
PS C:\Sena\Materias5Trimestre\Docker> docker run --rm -it busybox ping -c 2 servidor

ping: bad address 'servidor'
```

What's next:

Debug this container error with Gordon → `docker ai "help me fix this container error"`

```
PS C:\Sena\Materias5Trimestre\Docker>
```

2.9.práctica final del módulo: junta red + volumen + variables + exec.

Paso 1 — crear red y volumen

```
PS C:\Sena\Materias5Trimestre\Docker> docker network create red_demo
46d2be1871129e4104adb29c0fdc297fe9a55b6c1ef199ec1a50031fba64ebb6
PS C:\Sena\Materias5Trimestre\Docker> docker volume create vol_demo
vol_demo
PS C:\Sena\Materias5Trimestre\Docker>
```

Paso 2 — levantar el postgres

```
PS C:\Sena\Materias5Trimestre\Docker> docker rm -f db
db
PS C:\Sena\Materias5Trimestre\Docker> docker run -d --name db --network red_demo -v
vol_demo:/var/lib/postgresql/data -e POSTGRES_PASSWORD=demo postgres:16
b7db2b0cf9b8fd73722dc02ab96e18db43c1876503e4e41ed583988b18d60bd9
PS C:\Sena\Materias5Trimestre\Docker>
```


Paso 3 — entrar y crear la table

```
PS C:\Sena\Materias5Trimestre\Docker> docker exec -it db psql -U postgres
psql (16.14 (Debian 16.14-1.pgdg13+1))
Type "help" for help.
```

```
postgres=# CREATE TABLE notas (t text);
CREATE TABLE
postgres=# INSERT INTO notas VALUES ('sobrevivi');
postgres=# SELECT * FROM notas;
postgres=# \q
Use control-D to quit.
postgres=# █
```

Paso 4 — la prueba de persistencia

```
PS C:\Sena\Materias5Trimestre\Docker> docker exec -it db psql -U postgres -c "INSERT
    INTO notas VALUES ('sobrevivi');"
INSERT 0 1
```

What's next:

Try Docker Debug for seamless, persistent debugging tools in any container or image → `docker debug db`

Learn more at <https://docs.docker.com/go/debug-cli/>

```
PS C:\Sena\Materias5Trimestre\Docker>
```

Módulo 3 — Dockerfile

3.1 Estructura del proyecto: app.py y requirements.txt

```
app.py > ...
1  flask import Flask
2  = Flask(__name__)
3
4  @.route("/")
5  ∨ hola():
6  return "Hola desde mi imagen pro
7
8  .run(host="0.0.0.0", port=5000)
```

```
requirements.txt
1  flask
...
```

3.2 Contenido del Dockerfile

```
Dockerfile
1 FROM python:3.12-slim
2 WORKDIR /app
3 COPY requirements.txt .
4 RUN pip install -r requirements.txt
5 COPY . .
6 EXPOSE 5000
7 CMD ["python", "app.py"]
```

3.3 Construcción de la imagen con docker build

```
C:\Sena\Materias5Trimestre\Docke\mi-app3>docker build -t mi-app:1.0 .
[+] Building 14.6s (11/11) FINISHED          docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 185B 0.0s
=> [internal] load metadata for docker.io/library/python:3.12-slim 1.6s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/5] FROM docker.io/library/python:3.12-slim@sha256:78387bc3881b 9.0s
=> => resolve docker.io/library/python:3.12-slim@sha256:78387bc3881b 0.0s
=> => sha256:56235e4245636e401a28cf88944d543b29ee028ae3b 249B / 249B 0.1s
=> => sha256:1560cfed01dc4c72c07f789d860078d74466a 12.12MB / 12.12MB 8.5s
=> => sha256:dbd0b7849e6c06b61e1fa6b7ffe053f246ce007 4.27MB / 4.27MB 5.7s
=> => extracting sha256:dbd0b7849e6c06b61e1fa6b7ffe053f246ce00758906 0.1s
=> => extracting sha256:1560cfed01dc4c72c07f789d860078d74466a1f9a26b 0.3s
=> => extracting sha256:56235e4245636e401a28cf88944d543b29ee028ae3b6 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 430B 0.0s
=> [2/5] WORKDIR /app 0.1s
=> [3/5] COPY requirements.txt . 0.0s
=> [4/5] RUN pip install -r requirements.txt 2.8s
=> [5/5] COPY . . 0.0s
=> exporting to image 0.8s
=> => exporting layers 0.5s
=> => exporting manifest sha256:7bbf7150ef4a8948cdee377516776c790752 0.0s
=> => exporting config sha256:86a353d0c7be459727bc7405a120057b3a8e75 0.0s
=> => exporting attestation manifest sha256:d93d7572d6bc7ccc2d7e6d72 0.0s
=> => exporting manifest list sha256:932dcc84b475464c173427467f4557b 0.0s
=> => naming to docker.io/library/mi-app:1.0 0.0s
=> => unpacking to docker.io/library/mi-app:1.0 0.2s
```

```
C:\Sena\Materias5Trimestre\Docke\mi-app3>
```

3.4 Ejecución del contenedor app1 a partir de la imagen mi-app:1.0

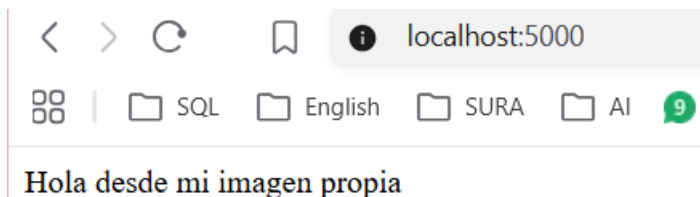
```
C:\Sena\Materias5Trimestre\Docker\mi-app3>docker run -d --name app1 -p 5000:5000 mi-app:1.0
df6e3ddb5d9007142d26b5cf03f606b23cd2ccc7161145d7d3c65a75b878d053
```

```
C:\Sena\Materias5Trimestre\Docker\mi-app3>docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	NAMES	STATUS
df6e3ddb5d90	mi-app:1.0	"python app.py"	5 seconds ago	app1	Up
e80f6da1f116	postgres:16	"docker-entrypoint.s..."	About an hour ago	db	Up
9b2a80dc053b	nginx	"/docker-entrypoint..."	2 hours ago	servidor	Up
408796df023f	nginx	"/docker-entrypoint..."	2 hours ago	api	Up
1a72e5f0e3fb	nginx	"/docker-entrypoint..."	2 hours ago	web	Up
16fa1488d42a	nginx	"/docker-entrypoint..."	2 hours ago	practical	Up

```
C:\Sena\Materias5Trimestre\Docker\mi-app3>curl localhost:5000
Hola desde mi imagen propia
C:\Sena\Materias5Trimestre\Docker\mi-app3>
```

3.5 Prueba de respuesta en localhost:5000



3.6 Archivo docker-compose.yml

```

docker-compose.yml
1  services:
2      web:
3          build: .
4          ports: ["8080:5000"]
5          environment: [DB_HOST=db]
6          depends_on: [db]
7      db:
8          image: postgres:16
9          environment: [POSTGRES_PASSWORD=secreto]
10         volumes: [datos:/var/lib/postgresql/data]
11
12  volumes:
13      datos:

```

3.7 Levantamiento del stack con docker compose up.

```

C:\Sena\Materias5Trimestre\Docker\mi-app3>docker compose logs -f web

C:\Sena\Materias5Trimestre\Docker\mi-app3>docker rm -f web
web

C:\Sena\Materias5Trimestre\Docker\mi-app3>docker compose up -d
[+] up 2/2
  ✓Container mi-app3-db-1  Running           0.0s
  ✓Container mi-app3-web-1 Started           0.1s

C:\Sena\Materias5Trimestre\Docker\mi-app3>docker compose ps
NAME                IMAGE                COMMAND              SERVICE    CREATED
STATUS             PORTS
mi-app3-db-1        postgres:16          "docker-entrypoint.s...  db         8 minutes a
go Up 8 minutes    5432/tcp
mi-app3-web-1        mi-app3-web          "python app.py"        web         8 minutes a
go Up 15 seconds   5000/tcp

C:\Sena\Materias5Trimestre\Docker\mi-app3>

```

3.8 Comandos de operación diaria con Docker Compose

```
#9 CACHED

#10 [4/5] RUN pip install -r requirements.txt
#10 CACHED

#11 [5/5] COPY . .
#11 DONE 0.0s

#12 exporting to image
#12 exporting layers 0.1s done
#12 exporting manifest sha256:006be6a16d484583ae77e6188cc0e6078062ce511f322e8ffbd124612dbd4a0c 0.0s done
#12 exporting config sha256:e24905c01c737488def6856f7548cff4d3c4e71c86e3703169d144fd8430771e 0.0s done
#12 exporting attestation manifest sha256:f4687e7f9361336d3cbfa0e66a8e8a71cd1251ef1dcd4b66e5bfff17316e6f069 0.0s done
#12 exporting manifest list sha256:e4f472f3adc708503925cfa78fb226845989a8d969ca3dedd1f8b08c58d9970f 0.0s done
#12 naming to docker.io/library/mi-app3-web:latest done
#12 unpacking to docker.io/library/mi-app3-web:latest 0.0s done
#12 DONE 0.2s

#13 resolving provenance for metadata file
#13 DONE 0.0s
[+] up 4/5
  ✓ Image mi-app3-web          Built          1.4s
  ✓ Network mi-app3_default    Created       0.0s
  ✓ Volume mi-app3_datos       Created       0.0s
  ✓ Container mi-app3-db-1     Started       0.5s
  - Container mi-app3-web-1    Starting      0.8s
Error response from daemon: failed to set up container networking: driver failed programming external connectivity on endpoint mi-app3-web-1 (d534ca6461f2e6ee4dabacea9a78e683595f0f8b6d357b9e4ba408c51699b0f4): Bind for 0.0.0.0:8080 failed: port is already allocated
```

What's next:

Debug this Compose error with Gordon → docker ai "help me fix this compose error"

```
C:\Sena\Materias5Trimestre\Docke\mi-app3>docker compose ps
NAME                IMAGE                COMMAND                  SERVICE    CREATED
mi-app3-db-1        postgres:16          "docker-entrypoint.s..." db          14 seconds ago
Up 14 seconds      5432/tcp
```

```
C:\Sena\Materias5Trimestre\Docke\mi-app3>
```

```
C:\Sena\Materias5Trimestre\Docker\mi-app3>docker compose restart web
[+] restart 0/1
  - Container mi-app3-web-1 Restarting 10.2s

C:\Sena\Materias5Trimestre\Docker\mi-app3>docker compose down
[+] down 3/3
  ✓ Container mi-app3-web-1 Removed 10.1s
  ✓ Container mi-app3-db-1 Removed 0.3s
  ✓ Network mi-app3_default Removed 0.2s

C:\Sena\Materias5Trimestre\Docker\mi-app3>|
```

Producto — Ejercicio integrador: MiniBlog en contenedores

P.1 + P.2 — Levantamiento del stack y mapeo de puertos

```

C:\Sena\Materias5Trimestre\Docke\miniblog>docker compose up -d --build
#1 [internal] load local bake definitions
#1 reading from stdin 535B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 185B done
#2 DONE 0.0s

#3 [auth] library/python:pull token for registry-1.docker.io
#3 DONE 0.0s

#4 [internal] load metadata for docker.io/library/python:3.12-slim
#4 DONE 0.8s

#5 [internal] load .dockerignore
#5 transferring context: 2B done
#5 DONE 0.0s

#6 [internal] load build context
#6 transferring context: 300B done
#6 DONE 0.0s

#7 [1/5] FROM docker.io/library/python:3.12-slim@sha256:78387bc3881b8273120a12e
be6c1ab22b018ccc2c9adf565ae1ac9b536e184ea
#7 resolve docker.io/library/python:3.12-slim@sha256:78387bc3881b8273120a12e
be6c1ab22b018ccc2c9adf565ae1ac9b536e184ea 0.0s done
#7 DONE 0.0s

#8 [3/5] COPY requirements.txt .
#8 CACHED

#9 [2/5] WORKDIR /app
#9 CACHED

#10 [4/5] RUN pip install -r requirements.txt
#10 CACHED

#11 [5/5] COPY . .
#11 CACHED

```

```

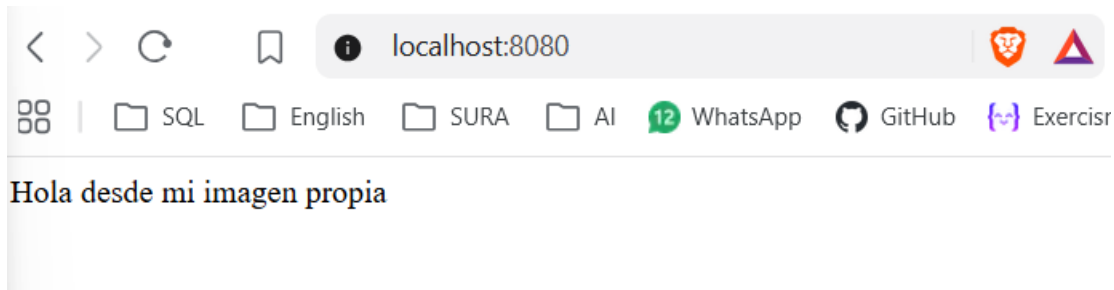
C:\Sena\Materias5Trimestre\Docke\miniblog>docker compose ps

```

NAME	STATUS	IMAGE	COMMAND	SERVICE	CREATED
miniblog-db-1	Up 12 minutes	postgres:16	"docker-entrypoint.s..."	db	12 minute
miniblog-web-1	Up 6 seconds	miniblog-web	"python app.py"	web	7 seconds

Ports: 5432/tcp (db), 0.0.0.0:8080->5000/tcp, [::]:8080->5000/tcp (web)

P.3 — Respuesta de la app en localhost:8080



P.4 y P.5 — Criterio 3: persistencia de los datos tras reiniciar el stack

```
C:\Sena\Materias5Trimestre\Docker\miniblog>docker compose exec db psql -U postgres -c "CREATE TABLE prueba(t text);"
CREATE TABLE

C:\Sena\Materias5Trimestre\Docker\miniblog>docker compose exec db psql -U postgres -c "INSERT INTO prueba VALUES ('miniblog');"
INSERT 0 1

C:\Sena\Materias5Trimestre\Docker\miniblog>docker compose down
[+] down 3/3
✓Container miniblog-web-1 Removed 10.3s
✓Container miniblog-db-1 Removed 0.2s
✓Network miniblog_default Removed 0.2s

C:\Sena\Materias5Trimestre\Docker\miniblog>docker compose up -d
[+] up 3/3
✓Network miniblog_default Created 0.0s
✓Container miniblog-db-1 Started 0.4s
✓Container miniblog-web-1 Started 0.5s

C:\Sena\Materias5Trimestre\Docker\miniblog>docker compose exec db psql -U postgres -c "SELECT * FROM prueba;"
 t
-----
 miniblog
(1 row)
```

P.7 Publicación del proyecto en GitHub.

```
C:\Sena\Materias5Trimestre\Docker\miniblog>git init
Initialized empty Git repository in C:/Sena/Materias5Trimestre/Docker/miniblog/.git/

C:\Sena\Materias5Trimestre\Docker\miniblog>git add .

C:\Sena\Materias5Trimestre\Docker\miniblog>git commit -m "Ejercicio integrador: MiniBlog en contenedores"
[main (root-commit) 52d39e6] Ejercicio integrador: MiniBlog en contenedores
 4 files changed, 29 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 app.py
 create mode 100644 docker-compose.yml
 create mode 100644 requirements.txt

C:\Sena\Materias5Trimestre\Docker\miniblog>git remote add origin https://github.com/marianauribe57-art/miniblog-docker.git

C:\Sena\Materias5Trimestre\Docker\miniblog>git branch -M main

C:\Sena\Materias5Trimestre\Docker\miniblog>git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 790 bytes | 790.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/marianauribe57-art/miniblog-docker.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

C:\Sena\Materias5Trimestre\Docker\miniblog>
```

git remote add origin https://github.com/marianauribe57-art/miniblog-docker.git

Módulo 5 — CI/CD con GitHub Actions. Como ya tienes el repo subido, esto encaja perfecto.

5.1 Estructura del workflow: archivo ci.yml

```
.github > workflows > ci.yml
You, hace 2 minutos | 1 author (You)
1  name: CI del miniblog
2  on: { push: { branches: [ main ] } }
3
4  jobs:
5    probar:
6      runs-on: ubuntu-latest
7      steps:
8        - uses: actions/checkout@v4
9        - uses: actions/setup-python@v5
10         with: { python-version: "3.12" }
11        - run: pip install -r requirements.txt
12        - run: python -m pytest -v
13
```

5.2 Ejecución del workflow en GitHub Actions

1 workflow run

Workflow ▾ Event ▾ Status ▾ Branch ▾ Actor ▾

✓ Agrega workflow de CI

CI del miniblog #1: Commit [8534381](#) pushed by [marianauribe57-art](#)

📅 2 minutes ago

🕒 16s main

...

5.3 Configuración de credenciales en GitHub Secrets

Secrets

Variables

Environment secrets

This environment has no secrets.

Manage environment secrets

Repository secrets

New repository secret

Name	Last updated	
DOCKER_TOKEN	now	
DOCKER_USER	1 minute ago	

5.4 Workflow de deploy: publicación de la imagen en Docker Hub

github.com/marianauribe57-art/miniblog-docker/actions

AI WhatsApp GitHub Exercism

ig-docker

Agents Actions Projects Wiki Security and quality Insights Settings

All workflows

Showing runs from all workflows

Filter workflow runs

3 workflow runs

	Workflow	Event	Status	Branch	Actor
✓	Agrega workflow de deploy a Docker Hub	CI del miniblog #2: Commit a0f169c pushed by marianauribe57-art	main	Sep 3, 5:00 PM GMT-5	...
✓	Agrega workflow de deploy a Docker Hub	Deploy del miniblog #1: Commit a0f169c pushed by marianauribe57-art	main	Sep 3, 5:00 PM GMT-5	...
✓	Agrega workflow de CI	CI del miniblog #1: Commit 8534381 pushed by marianauribe57-art	main	Sep 3, 4:38 PM GMT-5	...

Re-run triggered 6 minutes ago

marianauribe57-art

↻ a0f169c

main

Status

Success

Total duration

40s

Artifacts

—

deploy.yml

on: push

✓ probar9s

→

✓ desplegar22s

↺

—

+

Annotations

2 warnings

⚠️ probar

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, actions/setup-python@v5. For mo...

Show more

⚠️ desplegar

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v4, docker/login-action@v3. For mor...

Show more

5.5 Imagen publicada en Docker Hub



Mariana Uribe [Edit profile](#)

Community User Medellín Colombia

Repositories Starred

Search by repository name

Displaying 1 to 1 of 1 repositories

IMAGE

marianamumm/miniblog
marianamumm

3m 0